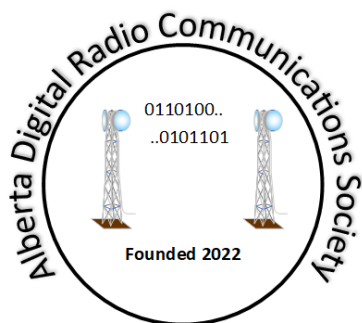




Digital Radio Projects

IP400 UFP Release Notes

Document Number: IP400-RLSE

Revision: 2.0

Status: Pre-Release

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Revision Status

Revision	Date	Description
1.0	March 24 th , 2025	Updated for phase 2 release
1.1	May 10 th , 2025	Added new MAC and IP addressing
1.2	May 15 th , 2025	Fixed data entry bugs and added revision numbering
1.3	May 30 th , 2025	Reworked FSM, added features, reworked menus
1.4	Nov, 2025	AX25 compatibility and KISS mode added
1.5	Jan, 2026	Memory allocation reworked
2.0	August, 2026	New UFP Platform

Table 1 Revision status

Reference Documents

Author	Issue Date	Document Number	Description
M. Alcock	Jan 2025	IP400-PHY	Physical Layer Specification
M. Alcock	Mar 2025	IP400-SPI	Radio Node SPI Protocol specification
M. Alcock	Mar 2025	IP400-CC2	Nucleo-CC2 Experimenter Node

Table 2 Reference Documents

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Introduction

The IP400 project was launched to experiment with digital mesh networking on the 400 MHz band, using commercial devices designed to run in this band.

To get the project rolling, code has been developed for an STM32WL33 microcontroller [2], on several platforms that support the basic mode. Further hardware will be developed in the future for different modes.

Operation of the node is segmented into a physical layer that runs on the microcontroller, and other layers that run on different host processors. The lower layer code contains the bare minimum to send and receive frames, repeating frames and building a mesh table, and contains a simple application.

The document describes the internal application which includes a simple setup menu, and the ability to change and store station and radio parameters, as well as a simple chat application to demonstrate the capabilities. It periodically sends a 'beacon' frame to build the mesh tables, which contains information about the station, including latitude, longitude and grid square. Provision has also been made to connect to a GPS receiver to update the position information dynamically.

The current revision level of the firmware is the first of the phase 2 to be released.

Roadmap

Figure 1 illustrates the roadmap for the IP400 node software.

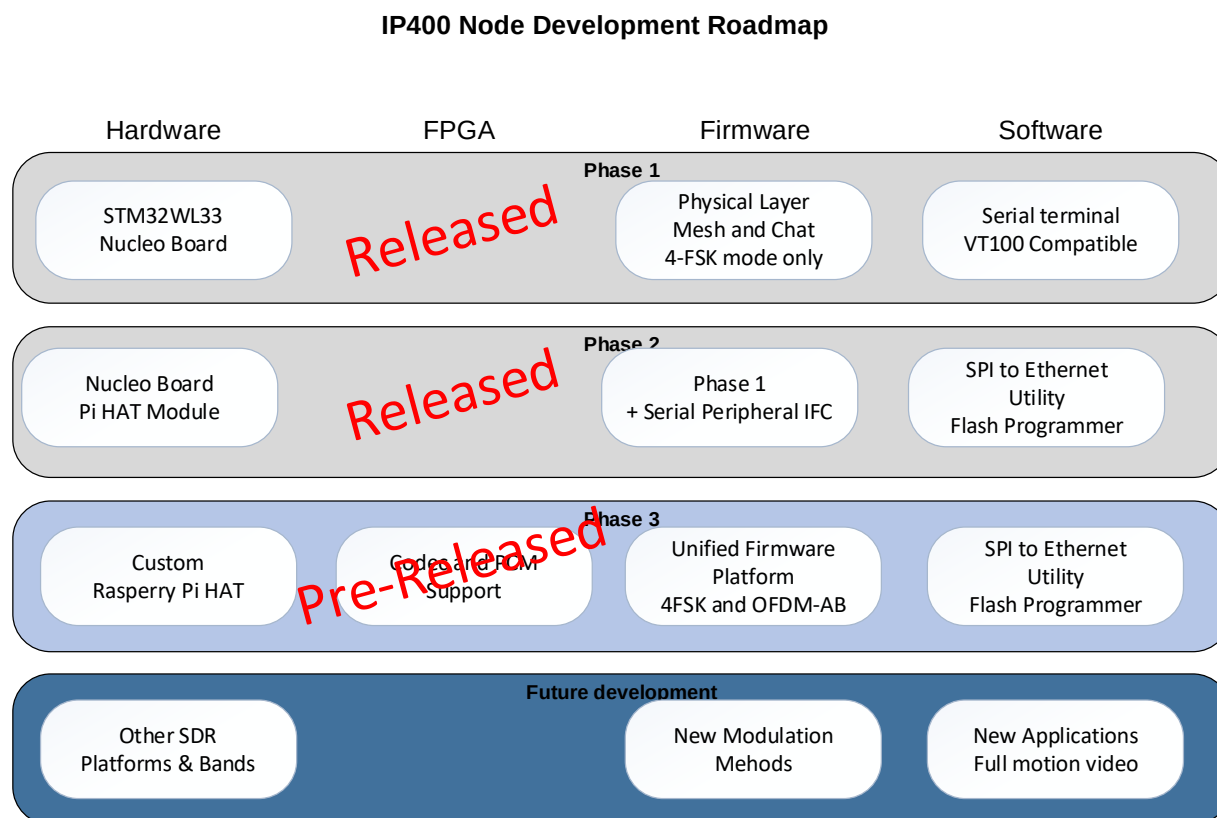


Figure 1 IP400 Node Development Roadmap

There are four phases of development for the IP400 node:

- Phase 1 is the initial evaluation phase using the Nucleo board and a simple chat application and is purely a firmware exercise. Using the PuTTY application, a connection can be made to the USART on the board. The software will implement a basic frame transmitter and receiver and be able to build a mesh table and repeat frames. The firmware can only be loaded using the ST integrated development environment (IDE) [3] and an integrated STLINK Debugger.
- Phase 2 utilizes either the Nucleo or an integrated HAT for the raspberry Pi platform. The firmware will be upgraded to include a high speed SPI, plus a downloader will be developed to alleviate the absolute requirement for the IDE, but support for an external STLINK debugger has been provided, so the IDE can also be utilized as an option. New applications on the Pi can be developed for routing, audio, file transfers, AX.25 and encapsulated IP.
- Phase 3 refactors the code into the Unified Firmware Platform, adds support for the Advanced Mesh Network Controller, and supports several modems, including OFDM-AB.

Phase 1 Release Notes

V0.1

Initial Feature Set

1. Simple frame transmitter using 4FSK modulation at 100 Kb/s.
2. Beacon frame generation.
3. Mesh table
4. Initial IP400 Frame format.
5. Setup menu

Phase 2 Release Notes

V1.0

New Features:

1. An experimental method of generating IP addresses has been added to mesh table.
2. Frames that are repeated now have a hop table that describes the routing methods.
3. SPI communication to host a processor has been added, details are available in a separate SPI specification document.
4. New code for the Pi to exchange data between the SPI and an ethernet port.
5. Wireshark dissector for packet examination using the generic dissector.
6. Allowances have been made for a local control packet to be utilized to change parameters such as frequency, bandwidth, etc, but not fully implemented.

Changes:

1. The previous scheme of using a port number in the callsign (MAC) address field was dropped, to be reconsidered in the future for multiple nodes. Callsigns are still compressed, the other bytes can be used for additional nodes under the same callsign, up to 4 bytes.
2. Packets are now processed based on their coding types. Further information is available in the SPI protocol specification document.
3. A message is sent to the GPS receiver, when enabled, to limit the type and frequency of message to reduce the overall traffic load.

Bugs Fixed:

1. A memory leak in the queuing code has been investigated and fixed.

V1.1

The major changes in this version are to address the same callsign on more than one node. A destination address can be one of the following:

1. Broadcast: All six bytes contain 'FF' in hex. Addresses all nodes that receive it.
2. Broadcast to all of the same callsign: compressed call sign + 'FFFF' in the unique ID field.
3. Individual address: compressed callsign + unique ID.

New Features:

1. The scheme in previous revision for IP addressing has been fully adopted.
2. New MAC addressing scheme was introduced: compressed callsign + unique ID.
3. Support for reuse of the same callsign in more than one unit has been added.
4. Chat mode was revised only to pick from the mesh table. Destination callsign selection logic has been revamped.

Changes:

1. Changed the conditional compilation for the second UAR/T to make it more generic for other purposes.
2. Created new IDE projects for the Pi HAT variants.
3. Added new hardware types in config.h, these are unique to each project.
4. Removed duplicate code in frame.h for sending frames, merged into one function.
5. Corrected LED test console output for Nucleo board.
6. Added ageing out of stations in the mesh table. After 30 minutes it goes into a 'lost' status, if heard again it will recover. After 60 minutes it is removed from the table.
7. Editing was applied to entries for the parameters to ensure correct case and format. Callsigns, for example are shifted to upper case if entered in lower.

V1.2

New Features:

To clear up any future confusion, the 'A' item on the menu now includes revision and build information
Firmware version: 1.2, Revision: nn Build Date: YYYY-MM-DD. The Revision matches a commit to the development repo, and the build date is the date and time of build, offset to Zulu time, and an interpreted date.

Bugs Fixed:

1. A bug was discovered in the setup for the beacon setting, which was overwriting the station capabilities. This bug has been rectified and limits on data entry have been added.
2. The backspace processing was also determined to not operate properly if it went back to the beginning of the line. This has also been fixed. The field validator now prints an error message if the validation fails.

V1.3

Bugs Fixed:

A bug was found that caused the transmitter to hang occasionally. The radio state machine was removed from its current source location and moved to a new file, and double buffering was added to increase overall throughput.

New Features:

1. Two new frame types; an echo request and response, were added and an echo mode (aka ping) was added to the chat mode. Depressing CTRL/E once enters into an 'on demand' echo mode, a second time sends it every 5 seconds. CTRL/E again will cancel it. A destination address has to be set first to use it, broadcast is not permitted.
2. The previous scheme of 'IP' addresses has been renamed to 'VPN', which more closely reflects what it is. No functional changes were introduced.

3. An AX.25 compatibility mode has been added, to enable backwards compatibility with older systems. This mode has to be enabled in the station menu.
4. A description field has been added to the setup menu, which is sent in beacon frames. The field can contain up to 32 characters.

V1.4

Changes:

1. A use case for the AX.25 compatibility mode has currently not been identified, so it has been moved to a compilation option.
2. The general sub menu return item is now 'Z'.
3. The 'Exit menu' item has been removed from the main menu as it was not functional.
4. Hitting the enter key on menu selection now repaints the current menu, this is useful when launching a terminal program.
5. The entry mode now shows the range of the entry required.

New Features:

1. The diagnostic items in the main menu have been moved to a new menu of their own, which includes the frame stats, GPS Echo mode (if enabled) and memory status.
2. Two new test modes have been added, a transmit CW mode that turns on the transmitter with no modulation, and a PRBS mode for spectral analysis.
3. An apply settings menu item has been added to the main menu, which reconfigures the radio based on current settings. No reset or write is required as long as power is applied. To make these changes permanent, they need to be written to flash.
4. The missing radio parameter for the PA drive mode has been added. See the relevant menu description for more details. Note: only the HP drive mode is connected on the Nucleo board, which limits the output to +14dBm.

V1.5

Bugs Fixed:

1. The memory management scheme has been reworked to properly make use of the heap space, and to report memory status properly. This was causing issues with larger frames.
2. The SPI code has been reworked to reassemble fragmented frames properly. The SPI buffer has been increased from 400 to 500 bytes, and in the case of fragmented frames, the reconstruction buffer is now allocated from the heap, instead of being static, and that memory has been re-distributed.
3. A bug in the transmitter that caused it to hang has been fixed.

New Features:

1. A KISS mode interface has been added that supports command TNC commands. The kiss mode can share the menu UAR/T or use the LPUART instead. In the latter mode the GPS receiver mode is disabled. Some additional KISS commands have been added, for more details see the appropriate document.
2. The buffers for the UART and LPUART have been increased to the maximum payload size.
3. The logger that was previously only used in one module has been removed, and the stream buffer definitions have been merged with the UAR/T.
4. The dynamic memory has been increased from 13 to 17K, memory allocation has been centralized and collects statistics on allocations on a module by module basis.
5. The memory allocation stats have been added to the memory status dump in the diagnostics status screen.
6. A set of SPI stats are now available in from the diagnostics menu, and the buffer usage has been added to the frame statistics. An ability to reset the frame and SPI stats has been added.
7. The SPI code has been reworked to handle its status correctly, and the scope toggle has been relocated to the interrupt service routine.
8. The use of the SPI slave select signal has been added, for better startup and synchronization. In the Nucleo, an additional jumper is required when connected to the Pi. In the mini-node, a hardware modification is required, refer to that document for more details.
9. Large data frames are now handled correctly. If a frame larger than the SPI buffer size of 500 bytes is encountered, but short enough to fit into one on-air frame, it is fragmented and concatenated for transmission. Frames larger than the on-air size are fragmented a second time and send as more than on on-air frame. The status and flag fields are updated accordingly to indicate the fragmentation status.
10. All four variants of node addressing have been implemented. It now recognizes broadcast, all call, address by call and VPN address, and, if enabled, the AX25 compatibility mode.

Phase 3 Release Notes

V2.0

New Features:

1. Complete refactoring of the Phase 2 code into the Phase 3 Unified Firmware Platform.
2. Separation of basic mesh code from modem code, conditional compilation for different platforms added.
3. Device independence added through vector table, allowing all modem types to share a common interface.
4. Addition of a buffer manager and extended frame length to 2048 bytes. Frames are packed together and sent when buffer is full, or a timeout occurs.
5. Refactored frame format to better accommodate IP frames up to 1500 bytes.
6. Callsign extension properly implemented and used throughout.
7. Hop table reduced to 2 entries to align with current AX.25 specification.
8. Refactored WL33 modem to operate in new code framework.
9. Addition of OFDM-AB modem.
10. Support for legacy nodes and new Advanced Mesh Network Controller.
11. Support for TLV320 codec either in standalone mode or with an FPGA.
12. FPGA downloader and insitu programming via primary or debug UART port.
13. DMA added to UART to improve throughput capabilities.
14. New diagnostics added for different platforms.